AMOD–A active building blocks on USB–C power

Jason Gallicchio and Claude, rev A, 2026–10–25 github.com/gallicchio/AMOD–and–RFMOD

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A5 Doppler
HB10D in socket
70 Hz per m/s
x101 audio

A4 LNA
slide up: on
IN < +–30 mV
x101, 1–30 M
keep gain > 12

40 to 0 dB
1M 100M

A3 x100
slide up: on
IN max +–40 mV
1.4 MHz wide

40 to 0 dB
10k 100M

A2 x10
slide up: on
IN max +–0.4 V
x10 high–Z
x5 into 50 ohm

20 to –20 dB
10k 100M

A1 Power
USB–C 5 V only
+5 V 1.5 A
–5 V 40 mA
+3.3 V 300 mA

A8 Heat
IO6 high: 0.8 W
HOT to touch
TEMP 10 mV/K
750 mV at 25 C

A7 PLL 4046
VCO -100 kHz
locks at x1
probe pins 9, 13

A6 1–bit ADC
IN 0–3.3 V
needs gateware
clock 1–10 MHz

A10 Impedance
Z = R X/(S–X)
DIP R: 47, 470,
4.7 k, 47 k

DRIVE +–4 V,
40 mA max
calibrate open
and short

A9 Mixer
RF, LO +–0.3 V
IF OUT is 1.5 k:
high–Z ADC,
short cable

A12 Audio
SPEAKER IN +–1 V
2.5 W into 4 ohm
MIC OUT x100
one–bit DAC on IO2

A11 Ultrasound
40 kHz only
TX 10 V p–p
from IO3, IO4
RX x100,
4 kHz high–pass
5.8 us per mm
there and back

github.com/gallicchio/
AMOD–and–RFMOD
handbook and generator

AMOD–A: active blocks on USB–C power. Each tile has a slide switch (up: on) and SMA jacks: cable DAC OUT to IN, OUT to ADC IN. The Pmod plug (left) takes the FPGA's lines; the handbook has the rest.

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Sheet:		
File: amod_a.kicad_pcb		
Title:		
Size: A4	Date:	**Rev:**
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